

```
// Define pins for HC-SR04
```

```
#define TRIG_PIN 9
```

```
#define ECHO_PIN 10
```

```
#define LED_PIN 13
```

```
void setup() {
```

```
    pinMode(TRIG_PIN, OUTPUT);
```

```
    pinMode(ECHO_PIN, INPUT);
```

```
    pinMode(LED_PIN, OUTPUT);
```

```
    Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
    // Send a 10 microsecond pulse to trigger
```

```
    digitalWrite(TRIG_PIN, LOW);
```

```
    delayMicroseconds(2);
```

```
    digitalWrite(TRIG_PIN, HIGH);
```

```
    delayMicroseconds(10);
```

```
    digitalWrite(TRIG_PIN, LOW);
```

```
    // Read echo time
```

```
    long duration = pulseIn(ECHO_PIN, HIGH);
```

```
    // Calculate distance in cm
```

```
    long distance = duration * 0.034 / 2;
```

```
    Serial.print("Distance: ");
```

```
Serial.print(distance);
```

```
Serial.println(" cm");
```

```
// Turn LED on if object is within 10 cm
```

```
if (distance <= 10) {
```

```
    digitalWrite(LED_PIN, HIGH);
```

```
} else {
```

```
    digitalWrite(LED_PIN, LOW);
```

```
}
```

```
delay(500); // Half-second delay for stability
```

```
}
```